SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 3 **Duration : 45 Minutes**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I S. LAKSHITA (192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot’s decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

CO1 – ASSESSMENT TOOL 2
Constraint-Based Problem Solving

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 3 Duration: 1 Hour **Total Marks: 30**

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Problem Understanding	25% (2.5)	2.5/2.5	2.5/2.5	2.4/2.5	7.4/7.5
Constraint Analysis	25% (2.5)	2.5/2.5	2.5/2.5	2.5/2.5	7.5/7.5
Solution Development	25% (2.5)	2.5/2.5	2.4/2.5	2.5/2.5	7.4/7.5
Application of Concepts	15% (1.5)	1.5/1.5	1.5/1.5	1.5/1.5	4.5/4.5
Justification	10% (1)	1/1	1/1	0.9/1	2.9/3
Total per Question	10 Marks	10/10	9.9/10	9.8/10	29.7/30

99

100

1.

Problem Formulation

The scheduling problem involves assigning three doctors (D1, D2, and D3) to three hospital shifts (Morning, Afternoon, and Night) while satisfying all specified constraints. Since the problem requires assigning values to variables under a set of restrictions, it is formulated as a Constraint Satisfaction Problem (CSP). The Backtracking Search algorithm is employed because it systematically explores feasible assignments and eliminates invalid solutions through constraint checking.

Problem Components

Variables

$$D = \{D_1, D_2, D_3\}$$

Domain

$$S = \{\text{Morning,Afternoon,Night}\}$$

Constraints

- $D_1 \neq$ Night
- $D_2 < D_3$(Morning < Afternoon < Night)
- Each shift must be assigned to exactly one doctor.
- $D_3 \neq$ Morning
- Every doctor must be assigned exactly one shift.

Constraint Analysis

The scheduling constraints consist of unary, binary, and global constraints.

The unary constraints restrict individual doctor assignments by preventing **D1** from working the Night shift and **D3** from working the Morning shift. The binary constraint requires **D2** to be scheduled before **D3**, thereby restricting the relative ordering of their assigned shifts. The global constraints ensure that no shift is assigned to more than one doctor and that each doctor is assigned exactly one shift.

Among these, the ordering constraint **D2 < D3** is the most restrictive because it directly influences multiple variable assignments and significantly reduces the feasible search space.

Backtracking Search Execution

The search begins with an empty assignment.

Doctor	Assigned Shift
D1	—
D2	—
D3	—

The algorithm explores assignments while verifying every constraint before proceeding further.

Assignment	Constraint Verification	Result
D1 = Morning	D1 is not assigned Night	Valid
D2 = Afternoon	Unique shift maintained	Valid
D3 = Night	D3 not in Morning, D2 precedes D3, unique assignment maintained	Valid

No constraint violation occurs during the search; therefore, no backtracking is required. Had any assignment violated a constraint, the algorithm would have returned to the previous state and explored the next available alternative.

Search Tree

Start

D1 = Morning

D2 = Afternoon

D3 = Night

Goal State Reached

Constraint Verification

Constraint	Verification	Status
D1 cannot work Night	D1 = Morning	Satisfied
D2 before D3	Afternoon < Night	Satisfied
Only one doctor per shift	Morning–D1, Afternoon–D2, Night–D3	Satisfied
D3 cannot work Morning	D3 = Night	Satisfied
Every doctor assigned exactly one shift	Verified	Satisfied

Final Schedule

Shift	Doctor
Morning	**D1**
Afternoon	**D2**
Night	**D3**

Application of AI Concepts

This scheduling problem represents a **Constraint Satisfaction Problem (CSP)** in Artificial Intelligence. The variables correspond to doctors, the domain consists of available shifts, and the constraints define the permissible assignments. Backtracking Search applies a depth-first exploration strategy with constraint checking at every stage, ensuring that invalid assignments are rejected immediately. This reduces unnecessary exploration and guarantees that the final solution satisfies all specified constraints.

Justification

Backtracking Search is the most suitable technique for this problem because it efficiently solves finite Constraint Satisfaction Problems by exploring only feasible assignments. Constraint checking after each assignment prevents unnecessary computation, while backtracking guarantees completeness by exploring alternative assignments whenever a violation occurs. Consequently, the algorithm produces a valid and optimal schedule that satisfies every scheduling constraint.

2.

Problem Formulation

A robot operates in a 5 × 5 grid environment and must travel from the **Start (S)** node to the **Goal (G)** node. The robot is permitted to move only in the four cardinal directions (Up, Down, Left, and Right). Obstacle cells cannot be traversed, and every movement has a uniform cost of **1**. The objective is to determine the shortest path using the Breadth-First Search (BFS) algorithm.

Assumed Grid

S ○ ○ X ○
○ X ○ X ○
○ ○ ○ ○ ○
X X ○ X ○
○ ○ ○ ○ G

Where:

- **S** = Start (0,0)
- **G** = Goal (4,4)
- **X** = Obstacle
- ○ = Free cell

Problem Components

Initial State: Start node (S)

Goal State: Goal node (G)

Operators:

- Move Up
- Move Down
- Move Left
- Move Right

Constraint Analysis

The robot can move only in four directions. Obstacle cells are inaccessible and therefore removed from the search space. Since every move has an identical cost, the problem represents an **unweighted graph**, making Breadth-First Search the most appropriate algorithm because it guarantees the shortest path.

Breadth-First Search Node Expansion

Iteration	Expanded Node	Queue After Expansion
1	S(0,0)	(0,1), (1,0)
2	(0,1)	(1,0), (0,2)
3	(1,0)	(0,2), (2,0)
4	(0,2)	(2,0), (1,2)
5	(2,0)	(1,2), (2,1)
6	(1,2)	(2,1), (2,2)
7	(2,1)	(2,2)
8	(2,2)	(2,3), (3,2)
9	(2,3)	(3,2), (2,4)
10	(3,2)	(2,4), (4,2)
11	(2,4)	(4,2), (3,4), (1,4)
12	(4,2)	(3,4), (1,4), (4,3), (4,1)
13	(3,4)	(1,4), (4,3), (4,1), G
14	**Goal (4,4)**	Goal Found

BFS Search Tree

```
              S
             /   \
         (0,1) (1,0)
           |       |
        (0,2)   (2,0)
           |       |
        (1,2)   (2,1)
             \    /
            (2,2)
           /     \
       (2,3)   (3,2)
          |         |
       (2,4)    (4,2)
          |
       (3,4)
          |
          G
```

Optimal Path

S
↓
(1,0)
↓
(2,0)
→
(2,1)
→
(2,2)
→
(2,3)
→
(2,4)
↓
(3,4)
↓
G

Path Cost

Number of moves = **8**

Cost per move = **1**

$$\boxed{\text{Total Cost} = 8}$$

Application of AI Concepts

The robot acts as an **intelligent agent** navigating an unweighted state-space graph. Breadth-First Search expands nodes level by level using a **FIFO queue**, ensuring that the first discovered path to the Goal is the shortest. Visited nodes prevent repeated exploration, while obstacle cells are excluded during node generation, making the search efficient and complete.

Justification

Breadth-First Search is the most suitable algorithm because every movement has the same cost. The algorithm explores all nodes at the current depth before proceeding to the next level, guaranteeing the shortest path to the Goal. It is complete, optimal for unweighted graphs, and effectively avoids invalid paths by ignoring obstacle cells during node expansion. Consequently, BFS satisfies all the specified constraints while producing the minimum-cost path.

3.

Problem Formulation

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment, where only adjacent rooms can be sensed. Some paths are blocked by obstacles, and risky zones impose an additional movement cost. The objective is to navigate safely from the Start (S) position to the Goal (G) while minimizing the total travel cost through intelligent decision-making.

(a) Constraint Analysis

The rescue robot must satisfy several operational constraints that directly influence its navigation strategy.

Constraint	Impact on Decision-Making	Priority
Movement allowed only in four directions	Restricts available actions at each state	High
Obstacles cannot be traversed	Invalid paths are eliminated from the search space	Very High
Limited sensing capability	Robot has incomplete knowledge and must explore dynamically	Very High
Each movement costs 1	Encourages shorter paths	High
Risky zones add an extra cost of 2	Safer routes may be preferred even if they are longer	Very High

Knowledge updated dynamically	Robot replans whenever new information becomes available	Very High

Constraint Prioritization

1. Avoid obstacles.
2. Minimize total path cost.
3. Avoid risky zones whenever possible.
4. Update knowledge using newly observed information.
5. Reach the survivor safely.

The most critical constraints are obstacle avoidance and minimizing total cost because they directly affect the robot's ability to complete the rescue mission successfully.

(b) Solution Approach Using Uniform Cost Search and Online Search

The rescue robot operates in a partially observable environment; therefore, an Online Search Agent continuously observes nearby rooms and updates its knowledge while navigating. For route selection, Uniform Cost Search (UCS) is used because movement costs are not uniform due to risky zones.

Solution Procedure

1. Initialize the robot at the Start position.
2. Observe all adjacent cells using onboard sensors.
3. Identify accessible, blocked, and risky cells.
4. Assign movement costs:
   - Normal cell = 1
   - Risky cell = 3 (1 movement cost + 2 additional risk cost)
5. Insert reachable states into the priority queue according to cumulative path cost.
6. Expand the node with the lowest cumulative cost.
7. Update the map whenever new neighbouring cells are discovered.
8. Ignore blocked cells and avoid risky zones whenever a lower-cost alternative exists.
9. Continue until the Goal node is reached.
10. Return the minimum-cost safe path.

Solution Flow

Start
|
Sense Adjacent Rooms
|
Detect Obstacles/Risky Zones
|
Update Knowledge Base
|
Calculate Path Cost

|

Select Lowest-Cost Node (UCS)

|

Move to Next Room

|

Goal Reached

(c) Application of AI Concepts

Intelligent Agent

The rescue robot functions as an intelligent agent by sensing its environment through sensors, processing the collected information, and selecting actions that maximize the probability of successfully rescuing the survivor while minimizing cost.

Rationality

The robot behaves rationally by selecting the action that minimizes cumulative travel cost while avoiding blocked and risky areas whenever possible. Every decision is based on the available information to maximize performance.

Environment Type

The disaster environment is characterized as:

Property	Description
Observability	Partially Observable
Determinism	Partially Deterministic
Dynamics	Dynamic
State Space	Discrete
Agent Type	Single-Agent

Because the robot can observe only adjacent rooms and conditions may change during navigation, it continuously updates its internal representation of the environment.

(d) Justification

Uniform Cost Search is the most suitable search strategy because movement costs are not identical throughout the environment. Unlike Breadth-First Search, which minimizes only the number of moves, Uniform Cost Search minimizes the **total accumulated cost**, including additional penalties associated with risky zones. The Online Search Agent further enhances performance by allowing the robot to update its knowledge dynamically whenever new information becomes available. This combination ensures safe navigation, optimal path selection, adaptability to changing conditions, and successful completion of the rescue task under all specified constraints.